TECHNICAL ASSESSMENT DOCUMENT

FairVia™ Technical Hypothesis Report

Material feasibility assessment for pre-commercial transition planning

APPLICATION dry goods packaging / shopping bag	MATERIAL TRANSITION PLA/PBAT blend
ASSESSMENT TYPE Technical Hypothesis	DATE 2026-06-10

01 EXECUTIVE SUMMARY

COMPATIBILITY LEVEL **MODERATE**

This assessment indicates MODERATE feasibility for the evaluated material transition within the current processing framework. Thermal (70) / Flow (85) / Mechanical (85) / Composite: 73 The system is operationally viable but thermally sensitive. Instability in thermal stability under processing conditions (score: 70/100) may influence material degradation risk and process reliability, particularly under elevated or variable processing temperatures. The target biodegradable material's narrow thermal window requires precision temperature control to prevent degradation onset during sustained production. Material cost variance is projected at +15–30%. Pilot validation is recommended with emphasis on temperature control precision and thermal distribution. Deployment Decision: CONDITIONAL GO

KEY RISK Variability in thermal stability under processing conditions (70/100) is the primary operational risk for this transition. This directly affects material degradation risk and process reliability and must be managed through rigorous control of temperature control precision and thermal distribution. Risk exposure increases proportionally under extended production cycles and elevated throughput conditions.

02 PROCESSING BEHAVIOUR

PROCESSING WINDOW	Processing window is operable but restricted by thermal stability under processing conditions (70/100). Sustained production requires tightly validated parameters; deviations outside the established range will cause measurable output instability and increased scrap rates.
THERMAL BEHAVIOUR	Marginal — processing temperature is near the material's degradation threshold (Thermal: 70/100). Active zone-by-zone temperature monitoring is required to prevent thermal degradation onset during extended production runs.
FLOW CHARACTERISTICS	Consistent — melt rheology within acceptable processing range (Flow: 85/100). Standard screw configuration and pressure settings are expected to maintain melt uniformity across production runs without active intervention.

03 PRODUCT REALIZATION

MECHANICAL BEHAVIOUR

Structural integrity of the final product is achievable under standard processing conditions (85/100). Mechanical performance meets commercial specification without formulation adjustment.

SURFACE QUALITY

Surface finish is expected to meet specification. Consistent melt flow (85/100) supports uniform surface formation under standard die and cooling conditions.

STRUCTURAL CONSISTENCY

Structural consistency is conditional on parameter control (Total: 73/100). Dimensional variation is expected at the margins of the processing window; tooling and cooling adjustments may be required.

ASSESSMENT SCOPE

All three parameters are evaluated under controlled pilot conditions. Real-world variability is expected to exceed laboratory baseline figures.

APPLICATION IMPLICATION

dry goods packaging / shopping bag is viable subject to process optimisation. Pilot-scale validation is required before commercial commitment. Constraint control measures must be implemented and verified prior to full deployment.

FAILURE ANALYSIS

PRIMARY RISK

Thermal Processing Constraint

Variability in thermal stability under processing conditions (70/100) is the primary operational risk for this transition. This directly affects material degradation risk and process reliability and must be managed through rigorous control of temperature control precision and thermal distribution. Risk exposure increases proportionally under extended production cycles and elevated throughput conditions.

SECONDARY RISK

Process Interaction Risk

Flow variability (85/100) interacts with mechanical performance (85/100), amplifying instability in process consistency under continuous production conditions. If flow parameters drift outside the validated tolerance window, mechanical behaviour is expected to degrade in tandem — compounding output variability beyond what the primary constraint alone predicts.

MECHANISM

polyethylene-based systems generally provide a wider melt-processing tolerance, lower moisture sensitivity, and broader residence-time stability, whereas the PLA/PBAT blend introduces a narrower processing window governed by PLA thermal sensitivity, PBAT-driven flexibility, blend morphology, and viscosity balance. Under dry goods packaging / shopping bag conditions, the resulting property mismatch concentrates around thermal stability under processing conditions. In film or sheet conversion, small changes in melt viscosity can translate directly into gauge variation, draw resonance, seal-window drift, surface non-uniformity, and web stability issues. As a result, material degradation risk and process reliability may move outside the qualified operating envelope unless temperature control precision and thermal distribution are validated through structured trials.

05 QUALITY ASSESSMENT

STABILITY

Moderate

Process stability index: 70/100. Conditional acceptance — enhanced in-line monitoring and SPC required.

CONSISTENCY

High

Flow consistency index: 85/100. Production consistency achievable within standard parameter tolerance.

EXPECTED DEVIATIONS

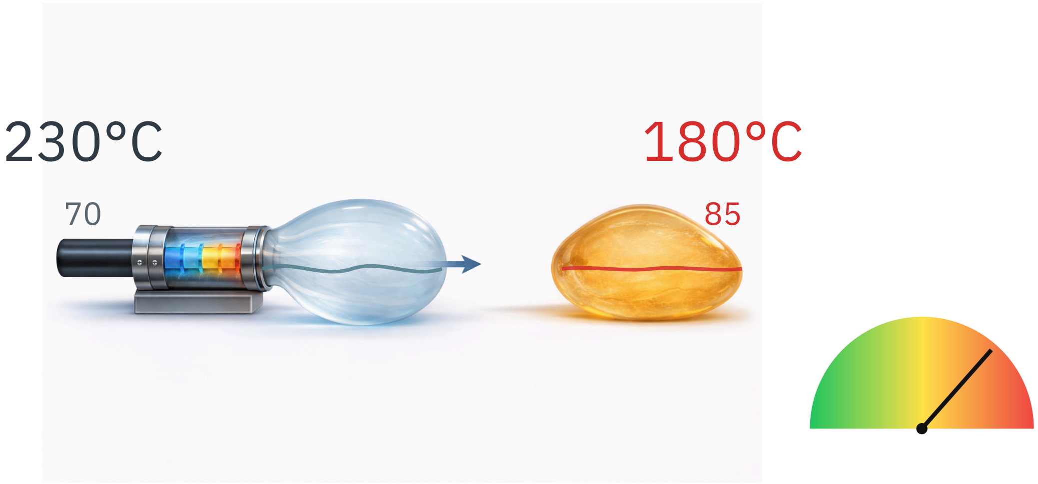
- Gauge or thickness variation of approximately ±8–12% may appear across the web width where melt pressure or draw stability drifts outside the qualified range
- Seal-window instability may occur if melt temperature, cooling rate, or blend morphology changes during extended production runs
- Surface haze, streaking, blocking tendency, or uneven gloss may increase during die start-up, high line-speed operation, or pressure fluctuation events
- Output consistency may decline during long runs if drying, residence time, and extrusion pressure are not controlled as validated parameters

PARAMETER PROFILE

THERMAL	70
FLOW	85
MECHANICAL	85

06 CONCEPTUAL PROCESSING COMPARISON

Illustrative comparison only. Temperature values and scores are conceptual indicators of relative processing tolerance, not recommended operating conditions.



07 RECOMMENDED NEXT STEP

Based on the MODERATE feasibility assessment (Total: 73/100), engineering validation targeting thermal performance is required before pilot approval. Implement control measures for temperature control precision and thermal distribution. Before pilot approval, confirm supplier technical evidence, define acceptable machine settings, and execute structured parameter trials to characterise the usable processing window. The validation plan should include start-up behaviour, steady-state operation, extended-run stability, and application-level output consistency. Re-evaluate system stability after stabilisation controls are confirmed, then proceed to pilot validation with defined acceptance criteria.

RECOMMENDED PATHWAY

01 Technical hypothesis screening

02 Engineering compatibility assessment

03 Controlled pilot validation

08 FAIRVIA SUPPORT SCOPE

FairVia supports material transition projects through a staged technical process, from early feasibility evaluation to pilot validation planning.

This report represents the first stage of that process. Where the identified risks are judged to be commercially relevant but technically manageable, the next phase should focus on engineering compatibility assessment and pilot preparation.

STAGE 01	STAGE 02	STAGE 03
Feasibility screening and technical hypothesis development	Engineering compatibility review and structured assessment	Pilot validation planning and implementation support

09 CLOSING NOTE

> *This report is based on a hypothesis-driven evaluation using the declared input parameters. All conclusions remain indicative until confirmed through physical trials and material testing.*
>
> *The findings in this report are intended to support structured technical decision-making prior to commercial implementation.*